

Q) $f(x) = x^3 - 3x^2 - 9$ $\frac{d}{dx}(x^3 - 3x^2 - 9) = (x^3)' - (3x^2)' - (9)'$ L1
 $f'(x) = 3x^2 - 6x$ $(x^3)' = 3x^2$ L2
 $f''(x) = 6x - 6$ $(3x^2)' = 6x$ L3

for maxima or minima, $0 = (x)'$

$f'(x) = 0$ $0 = 3x^2 - 6x$
 $\Rightarrow 3x^2 - 6x = 0$ $0 = (3-x)(x) \leftarrow$
 $\Rightarrow 3x(x-2) = 0$ $0 = (3-x)(x-0) \leftarrow$
 $\therefore x = 0, 2$ $x=3 \text{ not}$

when $x=0$, $f''(0) = -6 < 0$ $x=3 \text{ not}$

Thus the given function has a maxima at $x=0$ and the maximum value is -9 .

when $x=2$, $f''(2) = 6 > 0$ $x=3 \text{ not}$

Thus the given function has a minima at $x=2$ and the minimum value is -13 .

$$81 \quad f(x) = x^3 - 6x^2 + 9x + 5$$

$$f'(x) = 3x^2 - 12x + 9$$

$$f''(x) = 6x - 12$$

for maxima or minima,

$$f'(x) = 0$$

$$\Rightarrow 3x^2 - 12x + 9 = 0$$

$$\Rightarrow x(x-3) - 1(x-3) = 0$$

$$\Rightarrow (x-1)(x-3) = 0$$

$$\therefore x = 1, 3$$

$$\text{when } x = 1, \quad f''(1) = -6 < 0$$

Thus $f(x)$ has a maxima at $x = 1$ and the maximum value is 9.

$$\text{when } x = 3, \quad f''(3) = 6 > 0$$

Thus $f(x)$ has a minima at $x = 3$ and the minimum value is 5.

Q1 $f(x) = 2x^3 - 9x^2 + 12x - 3$

$$f'(x) = 6x^2 - 18x + 12$$

$$f''(x) = 12x - 18$$

for maxima or minima,

$$f'(x) = 0$$

$$\Rightarrow 6x^2 - 18x + 12 = 0$$

$$\Rightarrow x^2 - 3x + 2 = 0$$

$$\Rightarrow (x-2)(x-1) = 0$$

$$\therefore x = 1, 2$$

When $x=1$, $f''(1) = -6 < 0$

Thus $f(x)$ has a maxima at $x=1$ and the maximum value is 2.

When $x=2$, $f''(2) = 24 - 18 = 6 > 0$

Thus $f(x)$ has a minima at $x=2$ and the value is 1.

91 $f(x) = x^5 - 5x^4 + 5x^3 + 1$

$$f'(x) = 5x^4 - 20x^3 + 15x^2$$

$$f''(x) = 20x^3 - 60x^2 + 30x$$

for maxima or minima,

$$f'(x) = 0$$

$$5x^4 - 20x^3 + 15x^2 = 0$$

$$\Rightarrow x^2 (5x^2 - 20x + 15) = 0$$

$$\Rightarrow x^2 (x^2 - 4x + 3) = 0$$

$$\Rightarrow x^2 (x-3)(x-1) = 0$$

$$\therefore x = 0, 1, 3$$

when $x=0$, $f''(0) = 0$

$$\text{again, } f'''(x) = 60x^2 - 120x + 30$$

$$\text{when } x=0, f'''(0) = 30 \neq 0$$

Thus there is neither maxima nor minima at $x=0$.

$$\text{when } x=1, f''(1) = -10 < 0$$

Thus $f(x)$ has a maxima at $x=1$ and the maximum value is 0.

$$\text{when } x=3, f''(3) = 90 > 0$$

Thus $f(x)$ has minima at $x=3$ and the value is -28.

b) $f(x) = 5x^6 + 18x^5 + 15x^4 - 10$ (x) f

$$f'(x) = 30x^5 + 90x^4 + 60x^3$$

$$f''(x) = 150x^4 + 360x^3 + 180x^2$$

for maxima or minima, (x) f

$$f'(x) = 0$$

$$\Rightarrow 30x^5 + 90x^4 + 60x^3 = 0$$

$$\Rightarrow x^3(x^2 + 3x + 2) = 0$$

$$\Rightarrow x^3(x+1)(x+2) = 0$$

$$\Rightarrow x^3(x+1)(x+2) = 0$$

$$\therefore x = 0, -1, -2$$

when $x=0$, $f''(0) = 0$

$$\text{again } f'''(x) = 600x^3 + 1080x^2 + 360x$$

when $x=0$, $f'''(0) = 0$

Thus $f(x)$ has a maxima or minima at $x=0$.

$$\text{again } f^{IV}(x) = 1800x^2 + 2160x + 360$$

when $x=0$, $f^{IV}(0) = 360 > 0$

Thus $f(x)$ has a minima at $x=0$ and the value is -10 .

when $x=-1$, $f''(-1) = -30 < 0$

Thus $f(x)$ has a maxima at $x = -1$ and the value is -8 .

when $x = -2$, $f''(-2) = 24 > 0$

Thus $f(x)$ has a minima at $x = -2$ and the value is -26 .

2) $f(x) = x^3 - 3x^2 + 3x + 1$

$$f'(x) = 3x^2 - 6x + 3$$

$$f''(x) = 6x - 6$$

for maxima or minima,

$$f'(x) = 0$$

$$\Rightarrow 3x^2 - 6x + 3 = 0$$

$$\Rightarrow (x-1)^2 = 0$$

$$\Rightarrow x = 1$$

when $x = 1$, $f''(1) = 0$

again, if $f'''(x) = 6$

Thus the given function has neither maxima nor minima at $x = 1$.

20] $f(x) = x^3 - 6x^2 + 12x - 3$

$$f'(x) = 3x^2 - 12x + 12$$

$$f''(x) = 6x - 12$$

for maxima or minima,

$$f'(x) = 0$$

$$\Rightarrow 3x^2 - 12x + 12 = 0$$

$$\Rightarrow (x-2)^2 = 0$$

$$\therefore x = 2$$

$$\text{when } x=2, f''(2) = 0$$

$$\text{again, } f'''(x) = 6$$

Thus the given function has neither maximum nor minimum at $x=2$

22] $f(x) = \sin x (1 + \cos x)$

$$= \sin x + \frac{1}{2} \sin 2x$$

$$f'(x) = \cos x + \cos 2x$$

$$f''(x) = -(\sin x + 2 \sin 2x)$$

for maxima or minima,

$$f'(x) = 0$$

$$\Rightarrow \cos x + \cos 2x = 0$$

$$\Rightarrow \cos x + 2\cos^2 x - 1 = 0$$

$$\Rightarrow (\cos x + 1)(2\cos x - 1) = 0$$

$$\therefore \cos x = \frac{1}{2}$$

$$x = \frac{\pi}{3}$$

$$\text{or, } \cos x + 1 = 0$$

$$\Rightarrow \cos x = -1$$

$$\therefore x = (2n+1)\pi$$

According to the problem,

$$\text{when, } x = \frac{\pi}{3}, \quad f''\left(\frac{\pi}{3}\right) = -\left(\frac{\sqrt{3}}{2} + \sqrt{3}\right) \\ = -\frac{3\sqrt{3}}{2} < 0$$

Thus, the given function has a maximum value, for $x = \frac{\pi}{3}$ [proved]

$$\text{Sol } f(x) = \sin x + \cos 2x$$

$$f'(x) = \cos x - 2\sin 2x$$

$$f''(x) = -\sin x - 4\cos 2x$$

for maxima or minima,

$$f'(x) = 0$$

$$\Rightarrow \cos x - 2\sin 2x = 0$$

$$\Rightarrow \cos x - 4\sin x \cdot \cos x = 0$$

$$\Rightarrow \cos x (1 - 4\sin x) = 0$$

$$\therefore \cos x = 0$$

$$x = (2n+1)\frac{\pi}{2}$$

$$\text{or, } 1 - 4\sin x = 0$$

$$\sin x = \frac{1}{4}$$

$$x = \sin^{-1}\left(\frac{1}{4}\right)$$

$$\text{As } 0 < x < 2\pi,$$

$$\therefore x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\text{when } x = \frac{\pi}{2}, f''(x) = 3 > 0$$

Thus $f(x)$ has minima at $x = \frac{\pi}{2}$ and the minimum value is 0.

$$\text{when } x = \frac{3\pi}{2}, f''(x) = 5 > 0$$

Thus the given function has a minima at $x = \frac{3\pi}{2}$ and the minimum value is -2.

when $x = \sin^{-1}\left(\frac{1}{4}\right)$, $f''(x) = -\frac{1}{4} - 4\left(1 - 2 \cdot \frac{1}{16}\right)$
 $f''(x) = -\frac{30}{8} < 0$

Thus the given function has a maximum at $x = \sin^{-1}\left(\frac{1}{4}\right)$ and the maximum value is $\frac{9}{8}$.

28] $f(x) = t^3 - 3m^2t + 2n$

$f'(x) = 3t^2 - 3m^2$

$f''(x) = 6t$

for maxima or minima,

$f'(x) = 0$

$\Rightarrow 3t^2 - 3m^2 = 0$

$\therefore t = \pm m$

when $t = m$, $f''(m) = 6m > 0$

Thus $f(x)$ has a minima at $t = m$ and the value is $-2m^3 + 2n$.

when $t = -m$, $f''(-m) = -6m < 0$

Thus $f(x)$ has a maxima at $t = -m$ and the value is $2m^3 + 2n$.

(1) $y = x^2 + \frac{250}{x}$

$$y_1 = 2x - \frac{250}{x^2}$$

$$y_2 = 2 + 5 \times \frac{250}{x^3}$$

for maxima or minima

$$y_1 = 0$$

$$\Rightarrow 2x - \frac{250}{x^2} = 0$$

$$\Rightarrow x^3 = 125$$

$$\therefore x = 5$$

$$\text{when } x = 5, y_2 = 8 > 0$$

Thus the function has a minima at $x = 5$ and the minimum value is 75.

$$\text{ex) } f(x) = \frac{\ln x}{x}$$

$$f'(x) = \frac{\frac{x}{x} - \ln x}{x^2} = \frac{1 - \ln x}{x^2}$$

$$f''(x) = \frac{x^2 \cdot \frac{1}{x} (1 - \ln x) + (1 - \ln x) \cdot 2x}{x^4}$$

$$= \frac{x(1 - \ln x) + 2x(1 - \ln x)}{x^3} = \frac{3(1 - \ln x)}{x^3}$$

for maxima or minima,

$$f'(x) = 0$$

$$0 = \frac{1 - \ln x}{x^2}$$

$$\Rightarrow \frac{1 - \ln x}{x^2} = 0$$

$$0 = (1 - \ln x) \cdot \frac{1}{x^2}$$

$$\Rightarrow 1 - \ln x = 0$$

$$0 = 1 - \ln x$$

$$\Rightarrow \ln x = 1$$

$$1 = \ln x$$

$$\therefore x = e$$

$$\text{when } x = e, f''(e) = \frac{-e - 2e(1 - 1)}{e^3}$$

$$= \frac{-e}{e^3} = -\frac{1}{e^2} < 0$$

thus the given function has a maxima at $x = e$ and the maximum value is $\frac{1}{e}$

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if you solve [proved]

$$28] \quad y = \left(\frac{1}{x}\right)^x$$

$$\frac{x \ln x}{x} = (\infty) \text{ to } 1.55$$

$$\ln y = x \ln\left(\frac{1}{x}\right) = -x \ln x \quad \frac{x}{x} = (\infty) \text{ to } 1$$

$$\frac{dy}{dx} = -\left(\frac{1}{x}\right)^x (1 + \ln x) \quad \frac{1}{x} = (\infty) \text{ to } 1$$

$$\frac{d^2y}{dx^2} = -\frac{\left(\frac{1}{x}\right)^x}{x} + (1 + \ln x)^2 \left(\frac{1}{x}\right)^x$$

for maxima or minima,

$$\frac{dy}{dx} = 0$$

$$\text{or } \left(\frac{1}{x}\right)^x \neq 0$$

$$0 = \frac{x \ln x - 1}{x}$$

$$\Rightarrow \left(-\frac{1}{x}\right)^x (1 + \ln x) = 0$$

$$0 = x \ln x - 1$$

$$\therefore 1 + \ln x = 0$$

$$1 = x \ln x$$

$$\ln x = -1$$

$$1 = x$$

$$x = \frac{1}{e}$$

$$\text{when } x = \frac{1}{e}, \quad \frac{d^2y}{dx^2} = -\frac{\left(\frac{1}{e}\right)^{\frac{1}{e}}}{\frac{1}{e}} + 0 = -e^{\frac{1}{e}-1} < 0$$

Thus the given function has a

maximum at $x = \frac{1}{e}$ and the

maximum value is $e^{\frac{1}{e}}$

[proved]

20] $y = x^{-x}$ find $\frac{dy}{dx}$

$$\ln y = -x \ln x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = -1 - \ln x$$

$$\Rightarrow \frac{dy}{dx} = -y(1 + \ln x) = -x^{-x}(1 + \ln x)$$

$$\frac{d^2y}{dx^2} = -\frac{dy}{dx}(1 + \ln x) - \frac{y}{x}$$

for maxima $\frac{dy}{dx} = 0$

$$0 = -x^{-x}(1 + \ln x) = 0$$

$$\text{or } -x^{-x} \neq 0$$

$$\Rightarrow 1 + \ln x = 0$$

$$\Rightarrow \ln x = -1$$

$$\therefore x = \frac{1}{e}$$

when $x = \frac{1}{e}$, $\frac{d^2y}{dx^2} = -\frac{y}{x} = -\left(\frac{1}{e}\right)^{-\frac{1}{e}} < 0$

Thus the given function has a maximum at $x = \frac{1}{e}$ and the maximum value is $\left(\frac{1}{e}\right)^{-\frac{1}{e}}$

26) $f(x) = y = x^{1/x-1}$

$\ln y = \frac{1}{x} \ln x$

$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{1}{x^2} - \frac{\ln x}{x^2}$

$\frac{dy}{dx} = \frac{y}{x^2} (1 - \ln x) = x^{\frac{1}{x}-2} (1 - \ln x)$

$\frac{d^2y}{dx^2} = \frac{dy}{dx} \left(\frac{1}{x^2} - \frac{\ln x}{x^2} \right) - \frac{y}{x}$
 $= x^{\frac{1}{x}-2} (1 - \ln x)^2 - x^{\frac{1}{x}-3}$

for maxima $\frac{dy}{dx} = 0$

$\Rightarrow x^{\frac{1}{x}-2} (1 - \ln x) = 0$

$\Rightarrow \ln x = 1$

$\Rightarrow x = e$

when $x = e$, $\frac{d^2y}{dx^2} = 0 - e^{\frac{1}{e}-1} < 0$

Thus the given function has a maxima at $x = e$ and the maximum value is $e^{1/e}$

Q2) considering, one number is x

So the another number is $12-x$

Thus the product of two numbers

is, $y = x(12-x)$

$$y = 12x - x^2$$

$$\frac{dy}{dx} = 12 - 2x$$

$$\frac{d^2y}{dx^2} = -2 < 0$$

Thus the product of two numbers will be maximum when the numbers are equal.

for maximum,

$$\frac{dy}{dx} = 0$$

$$\Rightarrow 12 - 2x = 0$$

$$\therefore x = 6$$

$\therefore$ one number is 6 and

the other number is also 6.

Q6 considering, one number is x

So, the other number is $6-x$.

According to the problem,

$$y = x^2 + (6-x)^2$$

$$\Rightarrow y = x^2 + 36 - 2x + x^2$$

$$\Rightarrow y = 2x^2 - 2x + 36$$

$$\frac{dy}{dx} = 4x - 2$$

$$\text{Now } \frac{d^2y}{dx^2} = 4 > 0$$

Thus the sum of the number's squares will be minimum.

$$\text{for minimum, } \frac{dy}{dx} = 0$$

$$\Rightarrow 4x - 2 = 0$$

$$\therefore x = \frac{2}{4} = \frac{1}{2}$$

So one number is $\frac{1}{2}$ and

the other number is $\frac{11}{2}$

Q8] considering, one number is x
 other number is $100/x$

According to the problem,

$$y = x + \frac{100}{x}$$

$$\Rightarrow xy = x^2 + 100$$

$$\therefore y + \frac{dy}{dx} x = 2x$$

$$\Rightarrow \frac{dy}{dx} x = x - \frac{100}{x}$$

$$\Rightarrow \frac{dy}{dx} = 1 - \frac{100}{x^2}$$

$$\therefore \frac{d^2y}{dx^2} = \frac{200}{x^3}$$

for minima, $\frac{dy}{dx} = 0$

$$\Rightarrow 1 - \frac{100}{x^2} = 0$$

$$x^2 = 100$$

$$\therefore x = 10$$

when $x = 10$, $\frac{d^2y}{dx^2} = \frac{200}{10^3} = \frac{2}{10} > 0$

Thus the sum of the numbers will be least as possible when the number is 10.

Q1 Considering, One number x
Other number $20-x$

According to the problem,

$$y = x^3(20-x)$$

$$\Rightarrow y = 20x^3 - x^4$$

$$\therefore \frac{dy}{dx} = 60x^2 - 4x^3$$

$$\frac{d^2y}{dx^2} = 120x - 12x^2$$

for maxima or minima,

$$\frac{dy}{dx} = 0$$

$$\Rightarrow 60x^2 - 4x^3 = 0$$

$$\Rightarrow \frac{x^3}{x^2} = \frac{60}{4}$$

$\therefore x = 15$ and other number is 5

when $x = 15$, $\frac{d^2y}{dx^2} = 120(15) - 12(15)^2 = 1800 - 2700 = -900 < 0$

Thus, the product of the cube of one of them and other is maximum when the numbers are 15 and 5.

Let considering the length of the rectangle is x , the width is y and the given perimeter is p .

According to the problem,

$$p = 2(x + y)$$

$$\Rightarrow x + y = \frac{p}{2}$$

$$\therefore y = \frac{p}{2} - x \quad \text{--- (1)}$$

Area of the rectangle $A = xy$

$$= x \left(\frac{p}{2} - x \right)$$

$$= \frac{p}{2}x - x^2$$

$$\frac{dA}{dx} = \frac{p}{2} - 2x$$

$$\frac{d^2A}{dx^2} = -2 < 0$$

Thus the area will be maximum, when

$$\frac{dA}{dx} = 0$$

$$\Rightarrow \frac{p}{2} - 2x = 0$$

$$\therefore x = \frac{p}{4}$$

$$\text{and, } y = \frac{p}{2} - \frac{p}{4} = \frac{p}{4}$$

$$\therefore x = y = \frac{p}{4}$$

Hence, the largest rectangle with a given perimeter is a square.

Q99. considering the length and width of the rectangular area and y and, the radius of circle is r .
According to the problem,

$$4r^2 = x^2 + y^2$$

$$y^2 = 4r^2 - x^2 \quad \text{--- (1)}$$

$$(b+x) \cdot s = 9$$

$$\frac{9}{s} = b+x \quad \leftarrow$$

Area of the rectangle is, $x - \frac{9}{s} = b$ (1)

$$A = xy \quad \text{As } A \text{ is constant w.r.t } x \text{ and } y$$

$$\Rightarrow A = x \cdot \sqrt{4r^2 - x^2}$$

$$\Rightarrow A^2 = x^2 (4r^2 - x^2)$$

$$\Rightarrow Z = A^2 = 4r^2 x^2 - x^4$$

$$\frac{dZ}{dx} = 8xr^2 - 4x^3$$

$$\frac{d^2Z}{dx^2} = 8r^2 - 12x^2$$

for maxima, $\frac{dZ}{dx} = 0$

$$\Rightarrow 8xr^2 - 4x^3 = 0 \quad \frac{9}{s} = x \quad \leftarrow$$

$$\Rightarrow \frac{x^3}{x} = \frac{8r^2}{4} = 2r^2$$

$$x^2 = 2r^2$$

$$\Rightarrow x = r\sqrt{2}$$

from ① we get,

$$y^2 = 4r^2 - 2x^2$$

$$y^2 = 4r^2 - 2x^2$$

$$y = r\sqrt{2}$$

$$\therefore x = y = r\sqrt{2}$$

Thus the maximum rectangle inscribable in a circle is a square.

Let the side of the square be x .

$$2x = d$$

$$(x^2 + x^2) = d^2$$

$$2x^2 = d^2$$

$$x^2 = \frac{d^2}{2}$$

Let the side of the square be x .

$$x^2 + x^2 = d^2$$

$$2x^2 = d^2$$

$$x^2 = \frac{d^2}{2}$$

$$x = \frac{d}{\sqrt{2}}$$

$$52) u = f(x, y) = x^2 + 2y^2 - 4x + 4y - 3 \quad \text{--- (I)}$$

$$\frac{\partial f}{\partial x} = 2x - 4 \quad \text{--- (II)} \quad \text{and} \quad \frac{\partial f}{\partial y} = 4y + 4 \quad \text{--- (III)}$$

for maxima or minima,

$$\frac{\partial f}{\partial x} = 0$$

$$2x = 4$$

$$\therefore x = 2$$

$$\frac{\partial f}{\partial y} = 0$$

$$4y + 4 = 0$$

$$\therefore y = -1$$

Again diff (I) partially with respect to x and y ,

$$r = \frac{\partial^2 f}{\partial x^2} = 2$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 0$$

diff (III) partially with respect to y ,

$$t = \frac{\partial^2 f}{\partial y^2} = 4$$

Now,

$$rt - s^2 = 8 - 0 = 8 > 0$$

Thus the given function has a minima at $(2, -1)$ and the minimum value

is -9 .

Q2) $f(x, y) = x^3 + y^3 - 3x - 12y + 20$ — (i)

$$\frac{\partial f}{\partial x} = 3x^2 - 3 = 0 \quad \text{--- (ii)} \quad \frac{\partial f}{\partial y} = 3y^2 - 12 = 0 \quad \text{--- (iii)}$$

for maxima or minima, $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$

$$\frac{\partial f}{\partial x} = 0$$

$$\frac{\partial f}{\partial y} = 0$$

$$3x^2 - 3 = 0 \quad \Rightarrow \quad x = \pm 1$$

$$3y^2 - 12 = 0 \quad \Rightarrow \quad y = \pm 2$$

Again diff (ii) with respect to x and y respectively,

$$r = \frac{\partial^2 f}{\partial x \partial x} = 6x$$

Diff (iii) partially with respect to y ,

$$t = \frac{\partial^2 f}{\partial y \partial y} = 6y$$

when $x = 1$, $r = 6 > 0$

when $y = 2$, $t = 12 > 0$

$$rt - s^2 = 72 > 0$$

Thus the given function has a minima at $(1, 2)$ and the minimum value is 2.